Unit Test Cases

Test Case: Make sure that we are loading the data file correctly.

S. No.	Test Step	Test Data
1.	Read the data file	sample.csv
2.	Validate the number of records	record count = 9

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Unit Test Cases

Test Case: Make sure that we are loading the data file correctly.

S. No.	Test Step	Test Data
1.	Read the data file	sample.csv
2.	Validate the number of records	record count = 9

Test Case: Record Count by country is computed correctly

S. No.	Test Step	Test Data
1.	Read the data file	sample.csv
2.	Validate count by country	United Kingdom = 1 Canada = 2 United States = 4

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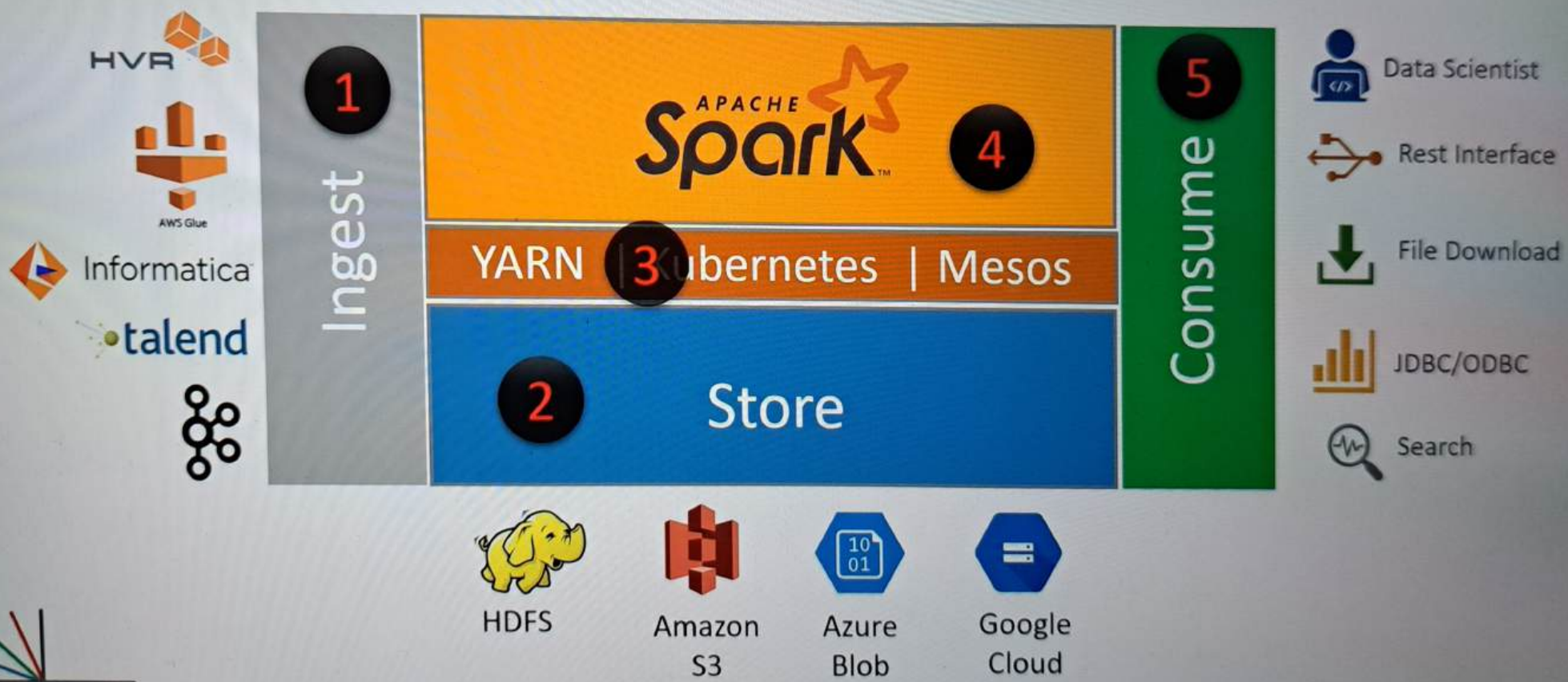
```python
    @classmethod
    def setUpClass(cls) -> None:
        cls.spark = SparkSession.builder \
            .master("local[3]") \
            .appName("HelloSparkTest") \
            .getOrCreate()

    def test_datafile_loading(self):
        sample_df = load_survey_df(self.spark, "data/sample.csv")
        result_count = sample_df.count()
        self.assertEqual(result_count, 9, "Record count should be 9")

    def test_country_count(self):
        sample_df = load_survey_df(self.spark, "data/sample.csv")
        count_list = count_by_country(sample_df).collect()
        count_dict = dict()
        for row in count_list:
            count_dict[row["Country"]] = row["count"]
        self.assertEqual(count_dict["United States"], 4, "Count for United States should be 4")
        self.assertEqual(count_dict["Canada"], 2, "Count for Canada should be 2")
        self.assertEqual(count_dict["United Kingdom"], 1, "Count for Unites Kingdom should be 1")

    # @classmethod
    # def tearDownClass(cls) -> None:
    #     cls.spark.stop()
```

Conception of Data Lake



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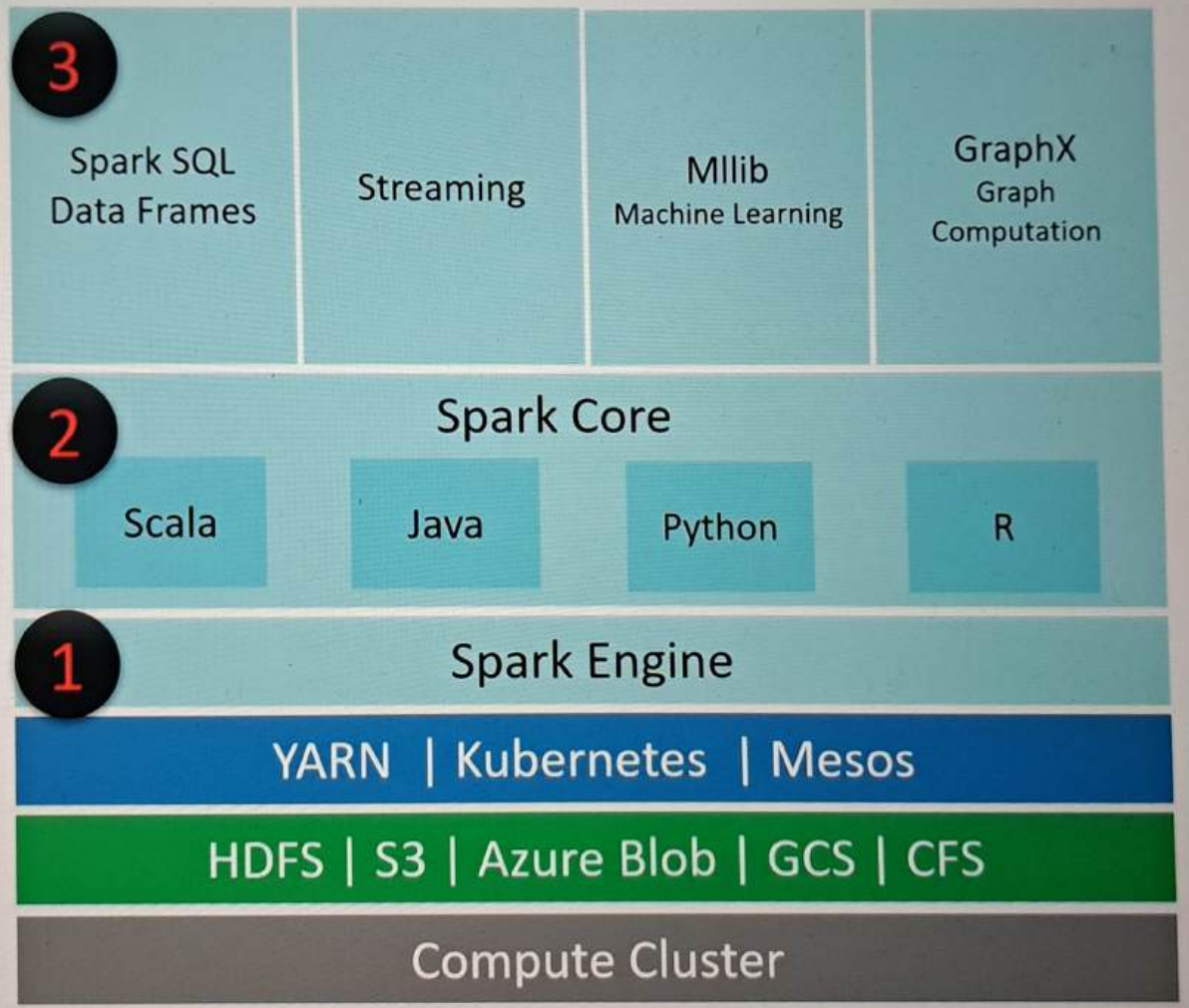
Spark Ecosystem



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Spark Installations

1. Local Mode – Command line REPL
2. Python IDE – PyCharm
3. Databricks Cloud – Notebooks
4. Other Notebooks – Jupyter Notebooks
5. Other Options – Cloud offerings

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Execution Methods

How to execute Spark Programs?

1. Interactive Clients
   spark-shell, Notebook
2. Submit Job
   spark-submit, Databricks Notebook, Rest API

Processing Model

Spark Client

Spark-submit

Spark Cluster

Cluster Manager (YARN)

A1 Driver

A1 Executor

A1 Executor

A1 Executor

A2 Driver

A2 Executor

A2 Executor

A2 Executor

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Spark Cluster Managers & Deployment Modes

Cluster Manager

1. local[n]
2. YARN
3. Kubernetes
4. Mesos
5. Standalone

Deployment Modes

1. Client Mode
2. Cluster Mode

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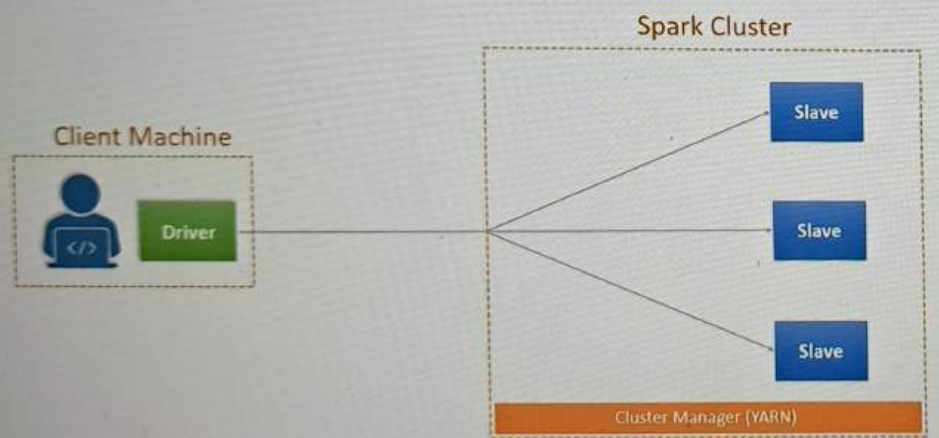
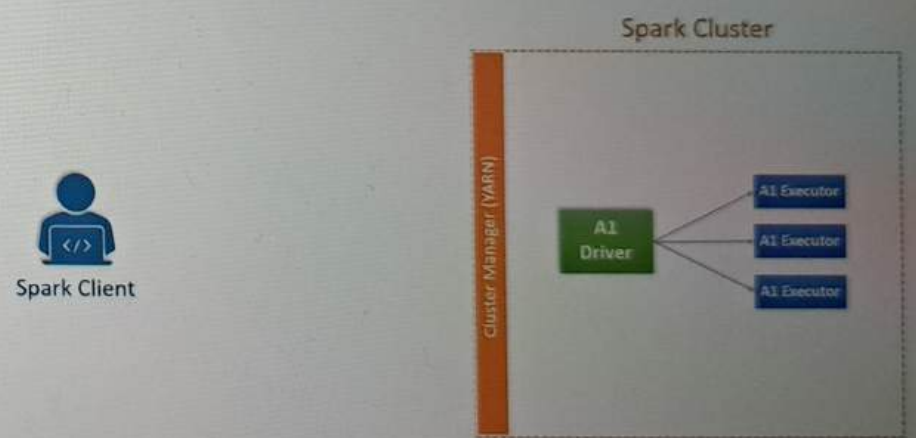
Deployment Modes

Client Mode



Cluster Mode



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Spark Execution Model

Cluster Managers	Execution Modes	Execution Tools
1. local[n]	1. Client	1. spark-shell
2. YARN	2. Cluster	2. spark-submit

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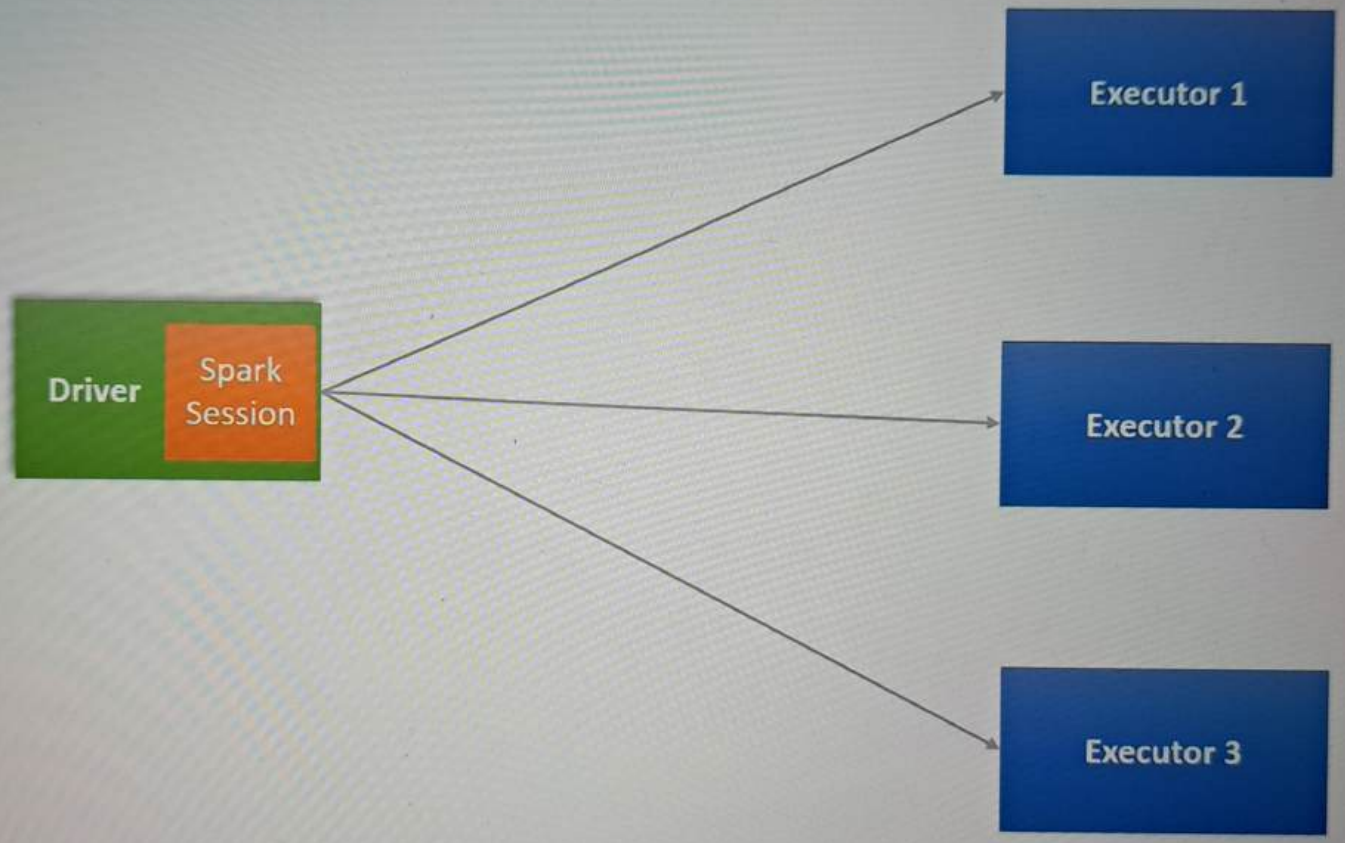
Developer Experience

1. Creating and Configuring Spark Project using your IDE
2. Configuring Log4J for your Spark Application
3. Creating and Configuring Spark Session
4. Managing your Spark Session Configurations using spark.conf
5. Creating a modular Structure for your Spark Application
6. Unit Testing Spark Application
7. Building and packaging your Spark Application
8. Deploying your Spark Application on a Cluster
9. Collecting Application Logs from Spark Cluster

Spark Session



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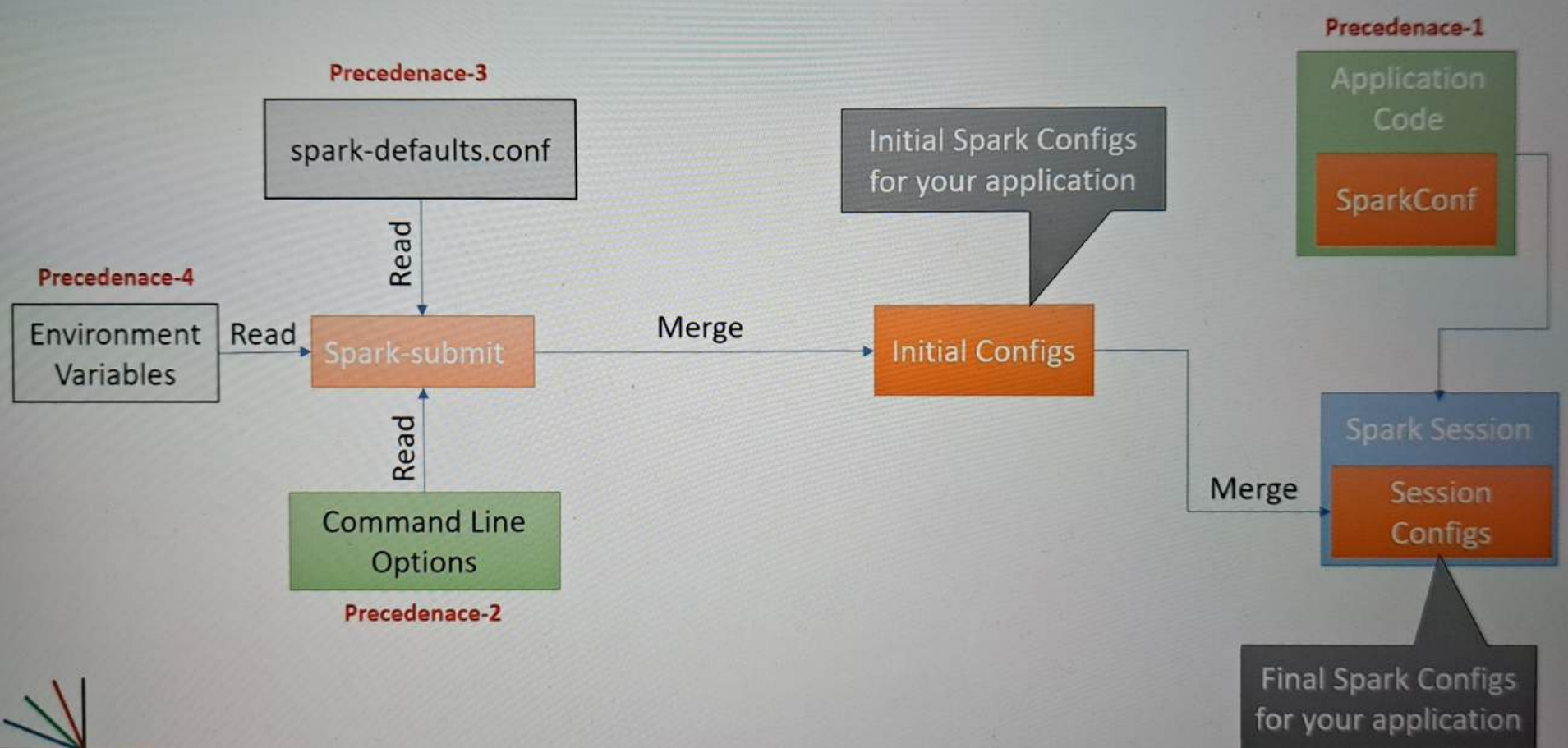
Spark Session Configs



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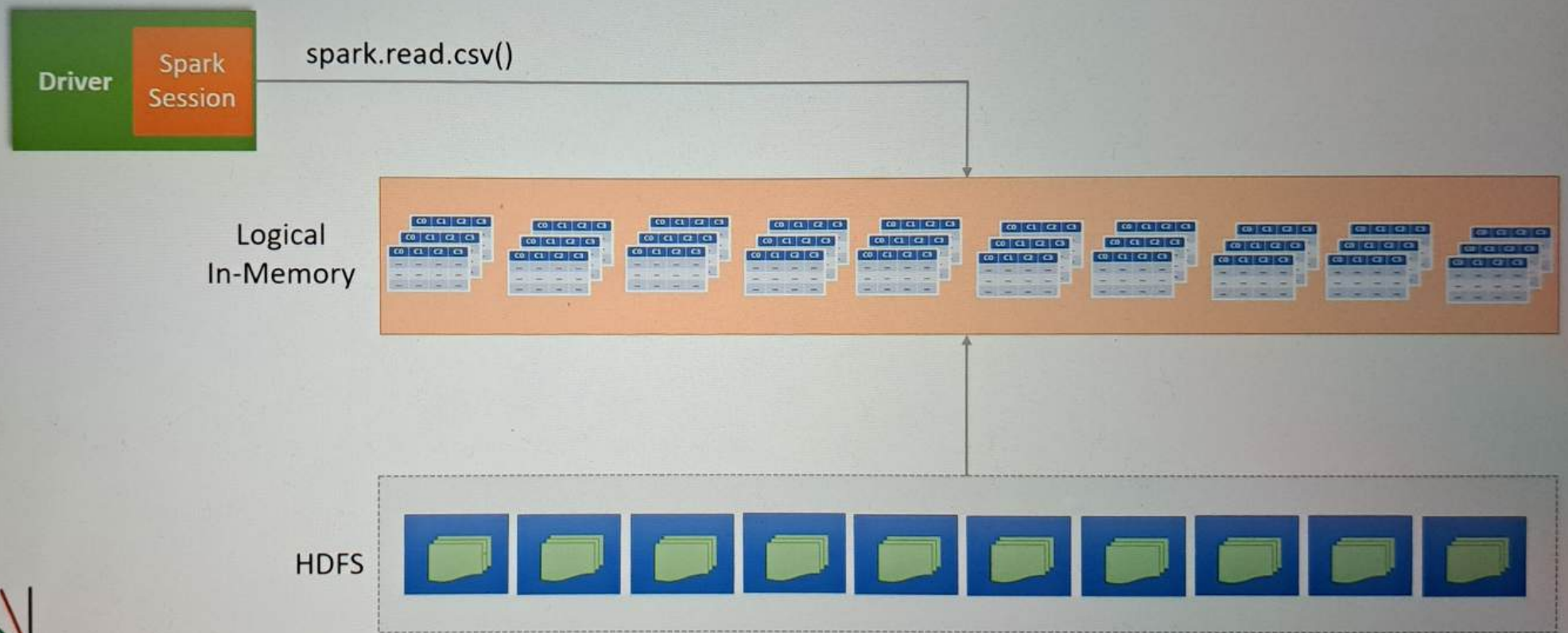
Spark Data Frame Partitions



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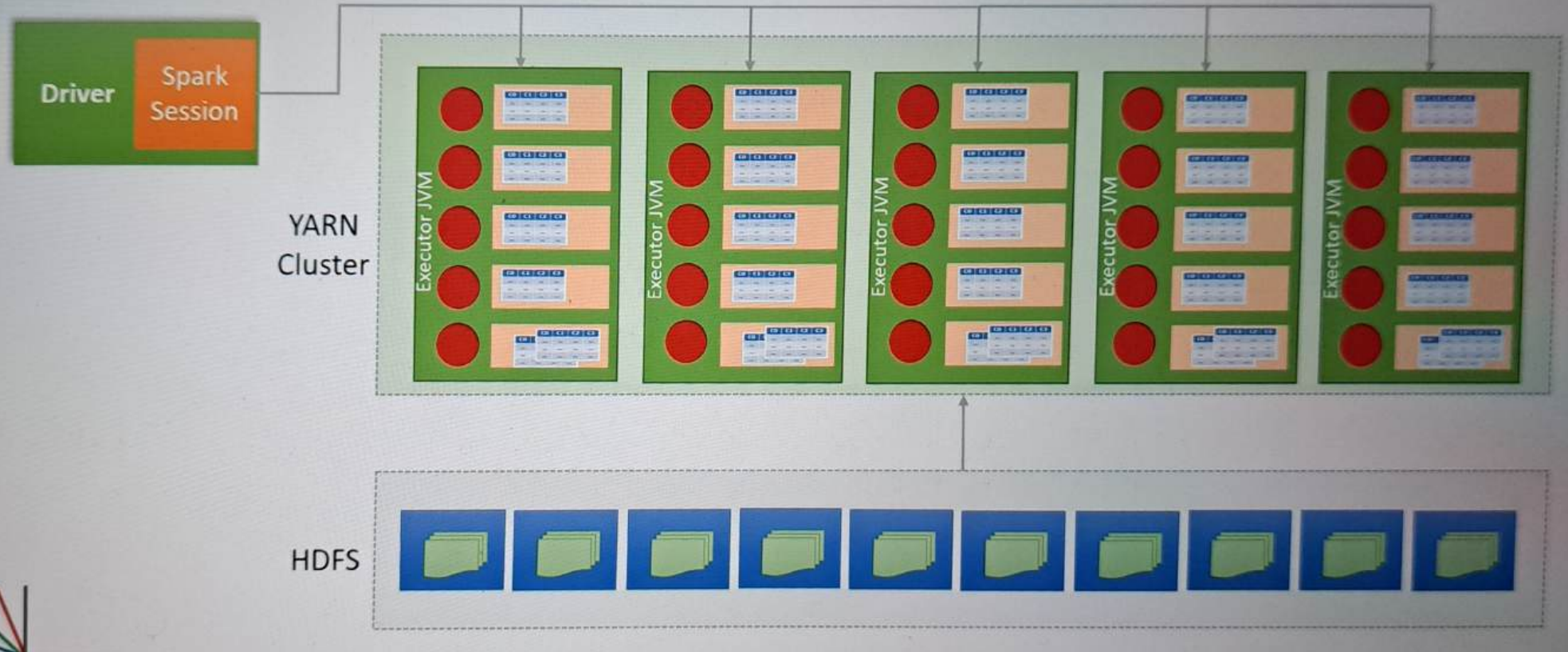
Spark Data Frame Partitions



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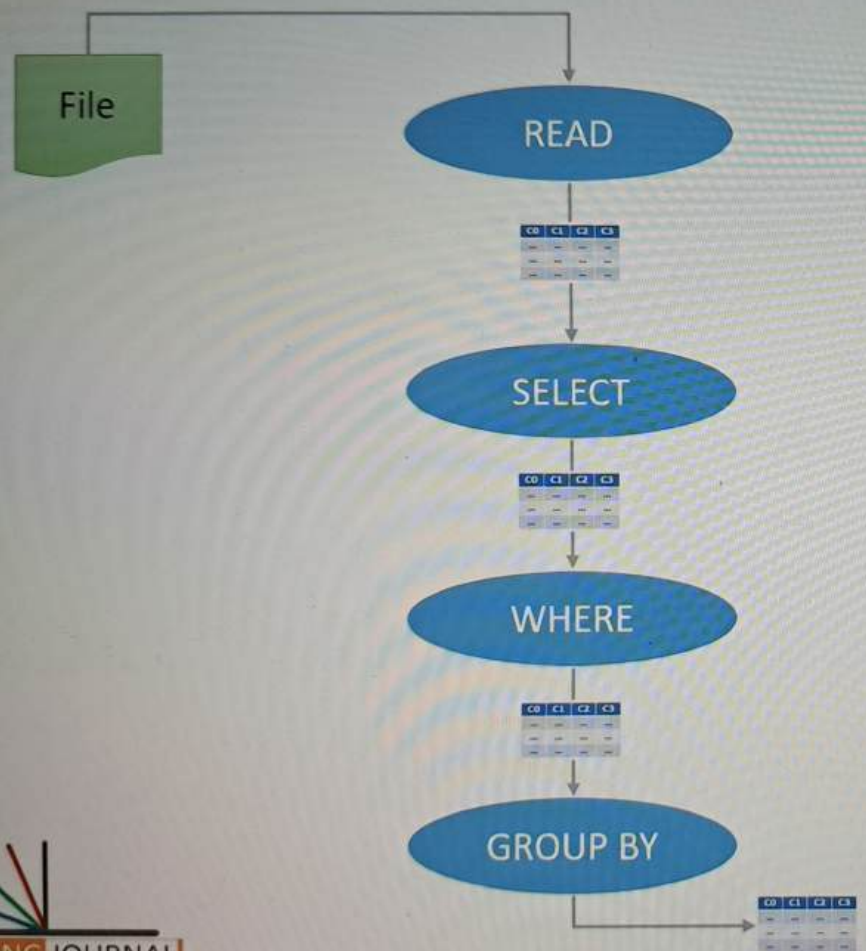
Spark Transformations DAG



1. Transformations
    1. Narrow Dependency
    2. Wide Dependency
2. Actions

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Wide Dependency Transformation

A transformation that requires data from other partitions to produce valid results.

Age	Gender	Country	state
37	Female	United States	IL
32	Male	Canada	
31	Male	United Kingdom	
31	Male	United States	TX
28	Female	Canada	
26	Female	United Kingdom	

Age	Gender	Country	state
37	Female	United States	IL
32	Male	Canada	
31	Male	United Kingdom	

Country	Age	Gender	state
United States	37	Female	IL
Canada	32	Male	
United Kingdom	31	Male	

Age	Gender	Country	state
31	Male	United States	TX
28	Female	Canada	
26	Female	United Kingdom	

Country	Age	Gender	state
United States	31	Male	TX
Canada	28	Female	
United Kingdom	26	Female	

shuffle/sort Exchange

Country	Age	Gender	state
United States	37	Female	IL
United States	31	Male	TX
Canada	28	Female	
Canada	32	Male	

Country	count
United States	2
Canada	2

Country	Age	Gender	state
United Kingdom	26	Female	
United Kingdom	31	Male	

Country	count
United Kingdom	2

Country	count
United States	2
United Kingdom	2
Canada	2

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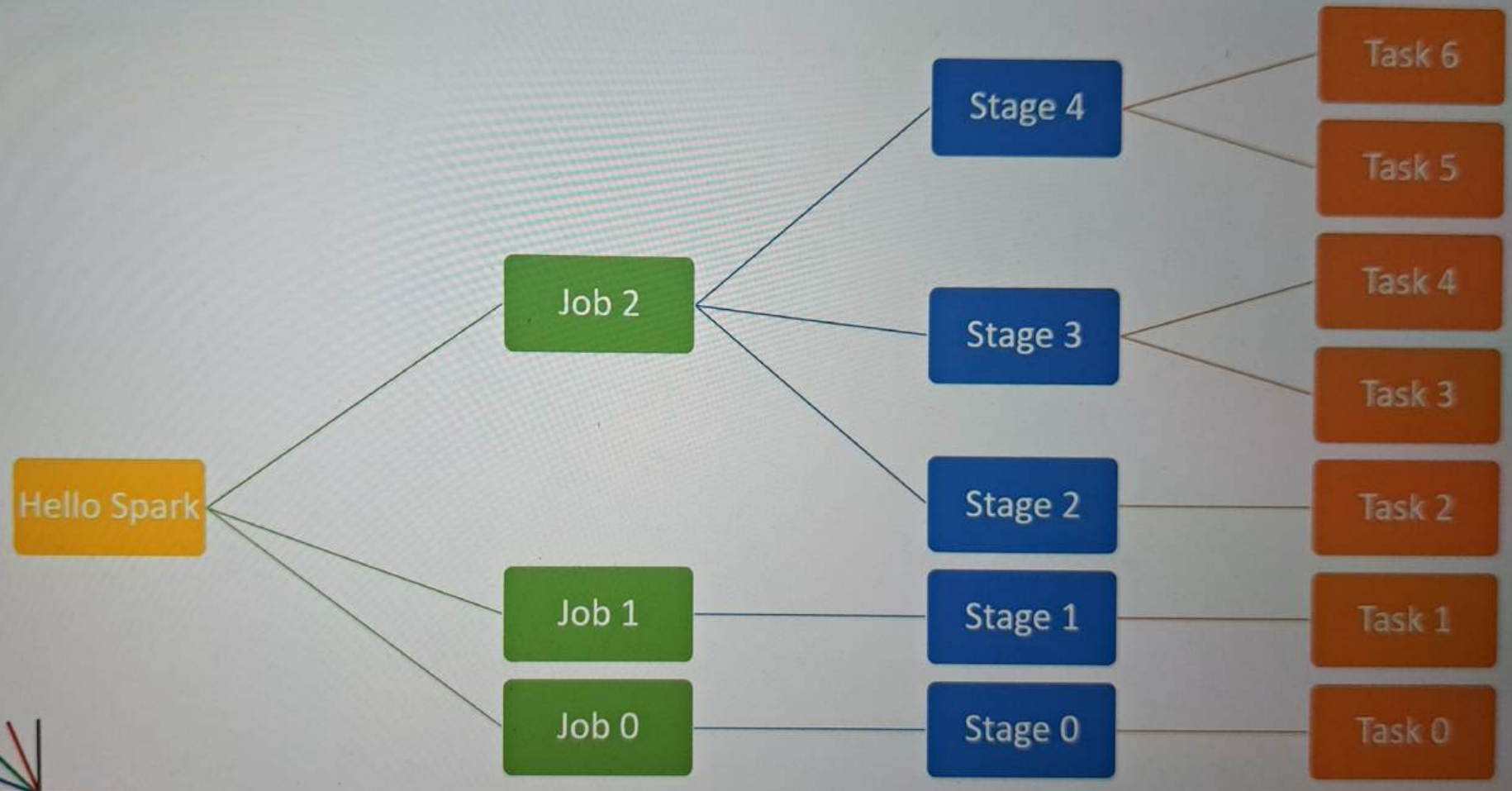
Spark Execution Plan



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Spark Execution Plan

```
survey_df = spark.read \
    .option("header", "true") \
    .option("inferSchema", "true") \
    .csv(data_file)
```

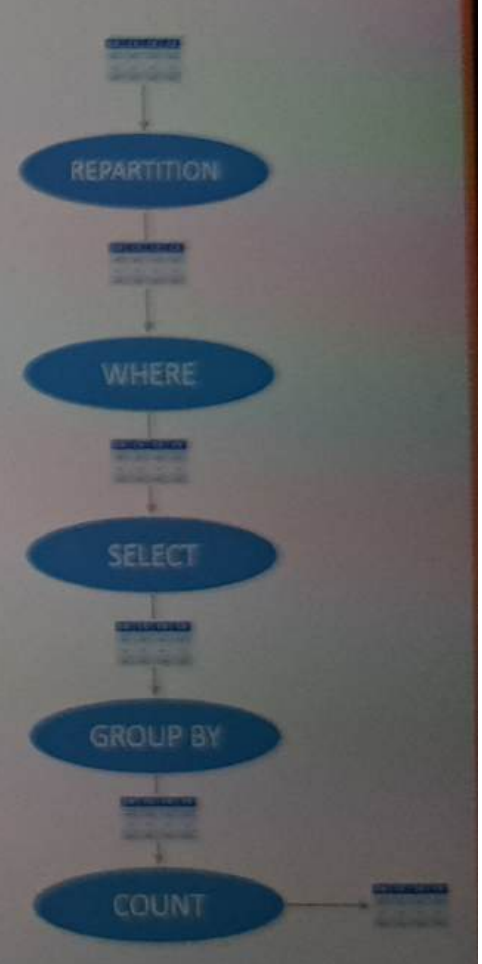
Job 0

Job 1

```
partitioned_survey_df = survey_df.repartition(2)
count_df = partitioned_survey_df.filter("Age < 40") \
    .select("Age", "Gender", "Country", "state") \
    .groupBy("Country") \
    .count()

logger.info(count_df.collect())
```

Job 2



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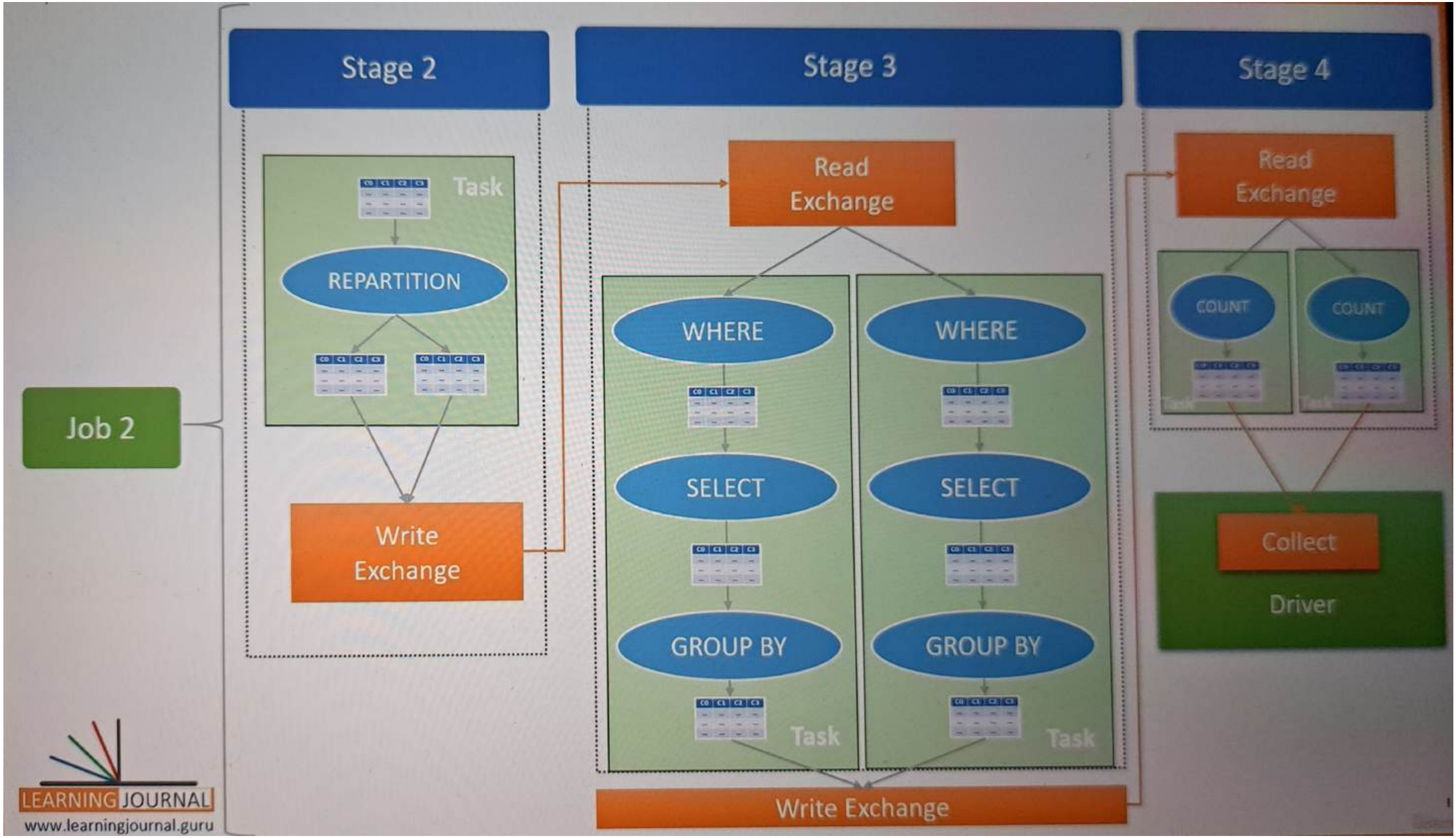
Job 2
Stage 2
Task
REPARTITION
Write Exchange
Stage 3
Read Exchange
WHERE
SELECT
GROUP BY
Task
WHERE
SELECT
GROUP BY
Task
Write Exchange
Stage 4
Read Exchange
COUNT
COUNT
Collect
Driver
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